

Both GW analysis weights can miss the true out-of-span projection

Candidate counterexample for arXiv:2507.09865

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Status. Candidate counterexample, likely valid. For two explicit uniform four-point metric spaces, both analysis algorithms in the source uniquely return the weight $(1/2, 1/2)$ for an out-of-span target, while the true GW projection is uniquely minimized at $(1/10, 9/10)$.

1 The source question

For fixed templates $\mathbb{X}^1, \dots, \mathbb{X}^S$, the source writes $\mathbb{Y}_\lambda$ for a GW barycenter and, for an arbitrary target $\mathbb{Y}$, considers

$$\mathbf{J}(\lambda) = \text{GW}^2(\mathbb{Y}_\lambda, \mathbb{Y}).$$

Supplement E asks whether, when $\mathbb{Y}$ is outside the GW barycenter span, the weights produced by its fixed-point and blow-up analysis algorithms coincide with, or at least lie near, a minimizer of $\mathbf{J}$.

APPENDIX E. THE ANALYSIS PROBLEM OUTSIDE THE BARYCENTER SPAN: A PRELIMINARY AND EMPIRICAL EXPLORATION

Assume that $\{\mathbb{X}^s\}_{s \in [S]}$ are fixed templates, and recall that we denote the elements of their GW barycenter span as $\mathbb{Y}_\lambda$, for $\lambda \in \Delta_{S-1}$. The objective of this section is to visualize, for an arbitrary target $\mathbb{Y}$, the heat map of the distance functional

$$\mathbf{J}(\lambda) := \text{GW}^2(\mathbb{Y}_\lambda, \mathbb{Y}), \tag{63}$$

on the simplex Δ_{S-1} , and the relation between its minima and the output of the analysis Algorithms 2 and 4 when they are applied to the target $\mathbb{Y}$.

By showing how such functional ‘looks like’ over the simplex of weights, we aim to provide preliminary experimental results to the following questions: *How the proposed GW analysis methods perform when the target shape $\mathbb{Y}$ does not belong to the GW barycenter space spanned by predetermined templates? What happens when the minimum of the above functional (63) is non-zero or cannot be attained exactly? Do the weights λ produced by the proposed GW analysis algorithms coincide with, or at least lie near, a minimizer of $\mathbf{J}$?*

The construction below gives a negative answer to the exact coincidence question, even for genuine finite metric-measure spaces and exact global GW couplings. It also exhibits a GW Karcher mean which is not a true barycenter.

2 Idea of the counterexample

The four-point metrics have a single nontrivial symmetry, the transposition of the first two points. Near a symmetric anchor, this symmetry acts on two perturbation coordinates by $(u, v) \mapsto (-u, -v)$. The true geodesic between the templates uses the identity alignment. The target is instead the midpoint obtained by aligning the second template with the transposition. Consequently, each source algorithm

sees an exact zero-residual midpoint and returns equal weights. But this midpoint lies on the wrong alignment branch. Projecting it onto the true barycenter geodesic gives a different weight.

Conditional negative definiteness of all metrics makes the GW coupling objective concave on the coupling polytope. Thus the full continuum of couplings reduces rigorously to the 24 permutations; no integrality assumption or numerical GW solver is used.

3 A four-point metric family

Give $\{1, 2, 3, 4\}$ the uniform probability vector $q = (1/4, 1/4, 1/4, 1/4)$ and, for $|u|, |v| \leq 1/5$, put

$$D(u, v) = \begin{pmatrix} 0 & 10 & 11+u & 12+v \\ 10 & 0 & 11-u & 12-v \\ 11+u & 11-u & 0 & 13 \\ 12+v & 12-v & 13 & 0 \end{pmatrix}. \quad (1)$$

All off-diagonal entries lie between $49/5$ and $66/5$, so every triangle inequality is strict.

Lemma 1. *For any two matrices in (1), their squared GW distance is the minimum, over permutations P of four letters, of*

$$\frac{1}{16} \|D(u, v) - PD(u', v')P^T\|_F^2.$$

Proof. Write $D(u, v) = 10(J - I) + E(u, v)$. The maximum absolute row sum of the symmetric matrix $E(u, v)$ is at most 7. Hence, for $x \perp \mathbf{1}$,

$$x^T D(u, v) x \leq -10\|x\|^2 + 7\|x\|^2 = -3\|x\|^2.$$

Thus every $D(u, v)$ is strictly conditionally negative definite.

For a coupling π with uniform marginals, the terms in the GW objective other than $-2 \operatorname{tr}(D(u, v)\pi D(u', v')\pi^T)$ are constant. If H has zero row and column sums, conditional negative definiteness gives

$$\operatorname{tr}(D(u, v)HD(u', v')H^T) \geq 0.$$

The GW objective is therefore concave on the coupling polytope. A concave function on this scaled Birkhoff polytope has a minimum at an extreme point, and those points are $P/4$. Substitution gives the formula. $\square$

Let S transpose points 1 and 2. The anchor $D(0, 0)$ has automorphism group exactly $\{I, S\}$: its unique edges of lengths 10 and 13 force the pairs $\{1, 2\}$ and $\{3, 4\}$, and the lengths 11 and 12 distinguish points 3 and 4. Hence a permutation outside $\{I, S\}$ changes some anchor edge by at least 1. The perturbations change an edge by at most $1/5$ on either side, so for such a permutation

$$\|D(u, v) - PD(u', v')P^T\|_F^2 \geq 2(1 - 2/5)^2 = 18/25. \quad (2)$$

For the remaining two permutations,

$$\|D(u, v) - D(u', v')\|_F^2 = 4((u - u')^2 + (v - v')^2), \quad (3)$$

$$\|D(u, v) - SD(u', v')S^T\|_F^2 = 4((u + u')^2 + (v + v')^2). \quad (4)$$

4 The templates and the false zero residual

Set

$$a_1 = (1/5, 0), \quad a_2 = (1/10, 1/5), \quad y = \frac{a_1 - a_2}{2} = (1/20, -1/10),$$

and let $\mathbb{X}_1, \mathbb{X}_2, \mathbb{Y}$ be the uniform metric spaces with matrices $D(a_1), D(a_2), D(y)$, respectively. Equations (2)–(4) give the following exact data.

pair	unique optimal permutation	squared Frobenius cost	GW ²
$(\mathbb{X}_1, \mathbb{X}_2)$	I	$1/5$	$1/80$
$(\mathbb{X}_1, \mathbb{Y})$	I	$13/100$	$13/1600$
$(\mathbb{X}_2, \mathbb{Y})$	S	$13/100$	$13/1600$

For a deterministic uniform coupling, the source's matrix $F(\mathbb{Y}, s)$ is just the optimally relabeled template. The blow-up representatives in its second method are the same matrices. Therefore both analysis quadratic programs use

$$D(a_1), \quad SD(a_2)S^T = D(-a_2), \quad D(y),$$

and

$$D(y) = \frac{1}{2}D(a_1) + \frac{1}{2}D(-a_2).$$

Both objectives have the unique zero at

$$\lambda_{\text{alg}} = (1/2, 1/2). \tag{5}$$

Thus both methods certify exact representability in their surrogate Karcher/fixed-alignment span.

The target is nevertheless outside the true GW barycenter span. Indeed, for weights $(1 - t, t)$ its synthesis cost is always

$$(1 - t) \text{GW}^2(\mathbb{X}_1, \mathbb{Y}) + t \text{GW}^2(\mathbb{X}_2, \mathbb{Y}) = 13/1600.$$

The unique optimal endpoint alignment is the identity. Its geodesic point

$$\mathbb{B}_t := D(a(t)), \quad a(t) = (1 - t)a_1 + ta_2 = (1/5 - t/10, t/5), \tag{6}$$

has synthesis cost $t(1 - t)/80 \leq 1/320 = 5/1600$. Hence $\mathbb{Y}$ is not a true barycenter for any weight.

For completeness, $\mathbb{B}_t$ is the unique barycenter class. Glue optimal couplings from any candidate $\mathbb{Z}$ to the two templates. The elementary identity

$$(1 - t)|r - z|^2 + t|s - z|^2 = t(1 - t)|r - s|^2 + |z - ((1 - t)r + ts)|^2$$

shows that equality in the two-point barycenter lower bound requires the induced template coupling to be endpoint-optimal and the candidate edge weights to equal the displayed convex combination almost everywhere. Since the endpoint coupling is uniquely the identity, this makes $\mathbb{Z}$ weakly isomorphic to (6).

5 The true projection weight

It remains to minimize $\mathbf{J}(t) = \text{GW}^2(\mathbb{B}_t, \mathbb{Y})$. The two permutations I, S have squared parameter costs

$$\begin{aligned} f_I(t) &= \|y - a(t)\|^2 = \frac{13}{400} + \frac{t}{100} + \frac{t^2}{20}, \\ f_S(t) &= \|y + a(t)\|^2 = \frac{29}{400} - \frac{9t}{100} + \frac{t^2}{20}. \end{aligned}$$

Both corresponding squared Frobenius costs are below the non-symmetry bound (2); Lemma 1 therefore gives

$$\mathbf{J}(t) = \frac{1}{4} \min\{f_I(t), f_S(t)\}.$$

The branches meet at $t = 2/5$. The first is increasing on $[0, 1]$, while the second has its unique minimum at $t = 9/10$. Direct evaluation yields

$$\min_{0 \leq t \leq 1} \mathbf{J}(t) = \mathbf{J}(9/10) = 1/125, \quad \mathbf{J}(1/2) = 1/100. \quad (7)$$

Consequently the true projection weight and the output of both analysis methods are

$$\lambda_{\text{proj}} = (1/10, 9/10) \neq (1/2, 1/2) = \lambda_{\text{alg}}.$$

The weight error is $2/5$ in either coordinate, and the algorithmic squared projection cost is $5/4$ times optimal. Multiplying every metric by a common constant preserves both weights and makes the additive gap in (7) arbitrarily large.

6 Scope

This completely answers the source supplement’s exact coincidence question in the negative and shows that an unconditional Karcher-mean/GW-barycenter equivalence is false. It does not preclude guarantees under additional alignment-consistency, curvature, separation, or uniqueness hypotheses, and it does not supply a full characterization of such hypotheses.

References

- [1] R. Díaz Martín, I. V. Medri, and J. M. Murphy, *Gromov–Wasserstein Barycenters: The Analysis Problem*, arXiv:2507.09865, 2025–2026 version.